MULTI-AREA IS-IS - Full Router Configuration

Level 1 + Level 2 | Wide Metrics | SPF | ECMP | Failover

Lab routers: Khalid-R1 - Khalid-R2 - Khalid-R3 - Khalid-R4 - Khalid-R5 - Khalid-R6

1. Addressing Summary

Router	Interface / Role	IP Address	Neighbor / Purpose
R1	Lo0	1.1.1.1/32	Area 49.0001 / L1
R1	Gi0/0	10.0.10.1/30	To R2 Gi0/0
R1	Gi1/0	192.168.1.1/24	PC-1 LAN
R2	Lo0	2.2.2.2/32	Area 49.0001 / L1-L2
R2	Gi0/0	10.0.10.2/30	To R1
R2	Gi1/0	10.0.20.1/30	To R3 / L2 cost 10
R2	Gi2/0	10.0.60.1/30	To R6 / L2 cost 40
R3	Lo0	3.3.3.3/32	Area 49.0002 / L2-only
R3	Gi0/0	10.0.20.2/30	To R2
R3	Gi1/0	10.0.30.1/30	To R4
R3	Gi2/0	10.0.50.1/30	To R6
R4	Lo0	4.4.4.4/32	Area 49.0003 / L1-L2
R4	Gi0/0	10.0.30.2/30	To R3
R4	Gi1/0	10.0.40.1/30	To R5
R4	Gi2/0	10.0.70.1/30	To R6
R5	Lo0	5.5.5.5/32	Area 49.0003 / L1
R5	Gi0/0	10.0.40.2/30	To R4
R5	Gi1/0	192.168.2.1/24	PC-3 LAN
R6	Lo0	6.6.6.6/32	Area 49.0004 / L1-L2
R6	Gi0/0	10.0.50.2/30	To R3
R6	Gi1/0	10.0.60.2/30	To R2
R6	Gi2/0	10.0.70.2/30	To R4
R6	Gi3/0	192.168.3.1/24	PC-2 LAN

Addressing is based on your latest GNS3 screenshot and CLI outputs. R5's local IS-IS tag was not supplied; this ready-to-paste reference uses KHALID-CORE-5. IS-IS process tags are local to each router.

The example is for building/rebuilding the lab. On an already configured router, compare the existing settings first; do not paste conflicting duplicate processes.

2. Khalid-R1 - Full Configuration

Area 49.0001 - Level 1 only

```
enable
configure terminal
hostname Khalid-R1
no ip domain lookup
ip cef
!
! ----- CREATE IS-IS PROCESS FIRST -----
router isis KHALID-CORE
 net 49.0001.0000.0000.0001.00
 is-type level-1
 metric-style wide
 exit
!
! ----- Loopback -----
interface Loopback0
 ip address 1.1.1.1 255.255.255.255
 ip router isis KHALID-CORE
 exit
!
! ----- Link to R2: Level 1 -----
interface GigabitEthernet0/0
 description TO-KHALID-R2
 ip address 10.0.10.1 255.255.255.252
 ip router isis KHALID-CORE
 isis circuit-type level-1
 no shutdown
 exit
!
! ----- PC-1 LAN -----
interface GigabitEthernet1/0
 description TO-PC1-LAN
 ip address 192.168.1.1 255.255.255.0
 ip router isis KHALID-CORE
 no shutdown
 exit
!
! ----- IS-IS Level 1 -----
end
write memory
```

3. Khalid-R2 - Full Configuration

Area 49.0001 - Level 1/2 boundary; metric tuning

```
enable
configure terminal
hostname Khalid-R2
no ip domain lookup
ip cef
!
! ----- CREATE IS-IS PROCESS FIRST -----
router isis HARCHAOUI-CORE
 net 49.0001.0000.0000.0002.00
 is-type level-1-2
 metric-style wide
 exit
!
interface Loopback0
 ip address 2.2.2.2 255.255.255.255
 ip router isis HARCHAOUI-CORE
 exit
!
! ----- Level 1: R1 -----
interface GigabitEthernet0/0
 description TO-KHALID-R1
 ip address 10.0.10.2 255.255.255.252
 ip router isis HARCHAOUI-CORE
 isis circuit-type level-1
 no shutdown
 exit
!
! ----- Level 2: R3 (preferred) -----
interface GigabitEthernet1/0
 description TO-KHALID-R3
 ip address 10.0.20.1 255.255.255.252
 ip router isis HARCHAOUI-CORE
 isis circuit-type level-2-only
 isis metric 10 level-2
 no shutdown
 exit
!
! ----- Level 2: R6 (higher-cost backup) -----
interface GigabitEthernet2/0
 description TO-KHALID-R6
 ip address 10.0.60.1 255.255.255.252
 ip router isis HARCHAOUI-CORE
 isis circuit-type level-2-only
 isis metric 40 level-2
 no shutdown
 exit
!
end
write memory
```

Gi1/0 uses cost 10 toward R3, while Gi2/0 uses cost 40 toward R6. With other backbone costs at 10, R2 prefers R3. If R3 fails, R6 remains a higher-cost alternative.

4. Khalid-R3 - Full Configuration

Area 49.0002 - Level 2 backbone router

```
enable
configure terminal
hostname Khalid-R3
no ip domain lookup
ip cef
!
! ----- CREATE IS-IS PROCESS FIRST -----
router isis HRCH-CORE
 net 49.0002.0000.0000.0003.00
 is-type level-2-only
 metric-style wide
 exit
!
interface Loopback0
 ip address 3.3.3.3 255.255.255.255
 ip router isis HRCH-CORE
 exit
!
interface GigabitEthernet0/0
 description TO-KHALID-R2
 ip address 10.0.20.2 255.255.255.252
 ip router isis HRCH-CORE
 isis circuit-type level-2-only
 no shutdown
 exit
!
interface GigabitEthernet1/0
 description TO-KHALID-R4
 ip address 10.0.30.1 255.255.255.252
 ip router isis HRCH-CORE
 isis circuit-type level-2-only
 no shutdown
 exit
!
interface GigabitEthernet2/0
 description TO-KHALID-R6
 ip address 10.0.50.1 255.255.255.252
 ip router isis HRCH-CORE
 isis circuit-type level-2-only
 no shutdown
 exit
!
end
write memory
```

5. Khalid-R4 - Full Configuration

Area 49.0003 - Level 1/2 boundary

```
enable
configure terminal
hostname Khalid-R4
no ip domain lookup
ip cef
!
! ----- CREATE IS-IS PROCESS FIRST -----
router isis KHAKID-CORE-4
 net 49.0003.0000.0000.0004.00
 is-type level-1-2
 metric-style wide
 exit
!
interface Loopback0
 ip address 4.4.4.4 255.255.255.255
 ip router isis KHAKID-CORE-4
 exit
!
! ----- Level 2: R3 -----
interface GigabitEthernet0/0
 description TO-KHALID-R3
 ip address 10.0.30.2 255.255.255.252
 ip router isis KHAKID-CORE-4
 isis circuit-type level-2-only
 no shutdown
 exit
!
! ----- Level 1: R5 -----
interface GigabitEthernet1/0
 description TO-KHALID-R5
 ip address 10.0.40.1 255.255.255.252
 ip router isis KHAKID-CORE-4
 isis circuit-type level-1
 no shutdown
 exit
!
! ----- Level 2: R6 -----
interface GigabitEthernet2/0
 description TO-KHALID-R6
 ip address 10.0.70.1 255.255.255.252
 ip router isis KHAKID-CORE-4
 isis circuit-type level-2-only
 no shutdown
 exit
!
end
write memory
```

6. Khalid-R5 - Full Configuration

Area 49.0003 - Level 1 only

```
enable
configure terminal
hostname Khalid-R5
no ip domain lookup
ip cef
!
! ----- CREATE IS-IS PROCESS FIRST -----
router isis KHALID-CORE-5
 net 49.0003.0000.0000.0005.00
 is-type level-1
 metric-style wide
 exit
!
interface Loopback0
 ip address 5.5.5.5 255.255.255.255
 ip router isis KHALID-CORE-5
 exit
!
! ----- Level 1: R4 -----
interface GigabitEthernet0/0
 description TO-KHALID-R4
 ip address 10.0.40.2 255.255.255.252
 ip router isis KHALID-CORE-5
 isis circuit-type level-1
 no shutdown
 exit
!
! ----- PC-3 LAN -----
interface GigabitEthernet1/0
 description TO-PC3-LAN
 ip address 192.168.2.1 255.255.255.0
 ip router isis KHALID-CORE-5
 no shutdown
 exit
!
end
write memory
```

KHALID-CORE-5 is a placeholder local process tag for a reproducible build, not a confirmed export from your existing R5. Change all three R5 occurrences together if your working tag differs.

7. Khalid-R6 - Full Configuration

Area 49.0004 - Level 1/2; Level 2 backbone links

```
enable
configure terminal
hostname Khalid-R6
no ip domain lookup
ip cef
!
! ----- CREATE IS-IS PROCESS FIRST -----
router isis KHALID-CORE-6
 net 49.0004.0000.0000.0006.00
 is-type level-1-2
 metric-style wide
 exit
!
interface Loopback0
 ip address 6.6.6.6 255.255.255.255
 ip router isis KHALID-CORE-6
 exit
!
! ----- Level 2: R3 -----
interface GigabitEthernet0/0
 description TO-KHALID-R3
 ip address 10.0.50.2 255.255.255.252
 ip router isis KHALID-CORE-6
 isis circuit-type level-2-only
 no shutdown
 exit
!
! ----- Level 2: R2 -----
interface GigabitEthernet1/0
 description TO-KHALID-R2
 ip address 10.0.60.2 255.255.255.252
 ip router isis KHALID-CORE-6
 isis circuit-type level-2-only
 no shutdown
 exit
!
! ----- Level 2: R4 -----
interface GigabitEthernet2/0
 description TO-KHALID-R4
 ip address 10.0.70.2 255.255.255.252
 ip router isis KHALID-CORE-6
 isis circuit-type level-2-only
 no shutdown
 exit
!
! ----- PC-2 LAN -----
interface GigabitEthernet3/0
 description TO-PC2-LAN
 ip address 192.168.3.1 255.255.255.0
 ip router isis KHALID-CORE-6
 no shutdown
 exit
!
end
write memory
```

8. VPCS Hosts - IP Configuration

```
! PC-1 (connected to R1)
ip 192.168.1.10/24 192.168.1.1
save

! PC-2 (connected to R6)
ip 192.168.3.10/24 192.168.3.1
save

! PC-3 (connected to R5)
ip 192.168.2.10/24 192.168.2.1
save
```

9. Verification Commands

```
! Cisco routers - in privileged EXEC mode
show isis neighbors
show isis database
show isis database detail
show isis topology
show ip route isis
show ip route 4.4.4.4
show clns interface GigabitEthernet2/0
show running-config interface GigabitEthernet2/0

! From PC-1 in VPCS
ping 192.168.2.10
ping 192.168.3.10
trace 192.168.2.10
trace 4.4.4.4
```

10. SPF Metric and Failover Experiment

Normal state: R2 prefers R2 -> R3 -> R4 (SPF metric 20 to R4) over R2 -> R6 -> R4 (cost 50). With R3 stopped in GNS3, the alternate path becomes R2 -> R6 -> R4. The route to R4 loopback may include another advertised prefix metric.

To return to equal-cost routing, set the R2 Gi2/0 Level 2 metric to 10. Expected R4 shortest-path cost: 20 via R3 and R6, provided other metrics are unchanged.

```
! Restore ECMP on R2 (optional experiment)
configure terminal
interface GigabitEthernet2/0
 isis metric 10 level-2
end
show isis topology

! Return to R3-preferred path
configure terminal
interface GigabitEthernet2/0
 isis metric 40 level-2
end
show isis topology
```

Reference fidelity: This PDF follows the organization of the supplied MPLS L3VPN Full Router Configuration PDF: addressing summary, full per-router copy-ready blocks, and final verification commands. R1 is grounded in the complete running-config you supplied; other router blocks are assembled from the topology and CLI evidence shared in the conversation. It is not an exact export of all six running configurations.